VARIATIONAL AUTO-ENCODER (VAE)

A Detailed Introduction · XU Chentianye

OUTLINE

This presentation is divided into 8 sections:

1. Autoencoder (AE)
2. The Idea of Variational Autoencoder (VAE)
3. Variational Lower Bound
4. Reparameterization Trick
5. Gaussian Distribution
6. Variational Lower Bound (Evidence Lower Bound)
7. VAE Network
8. References

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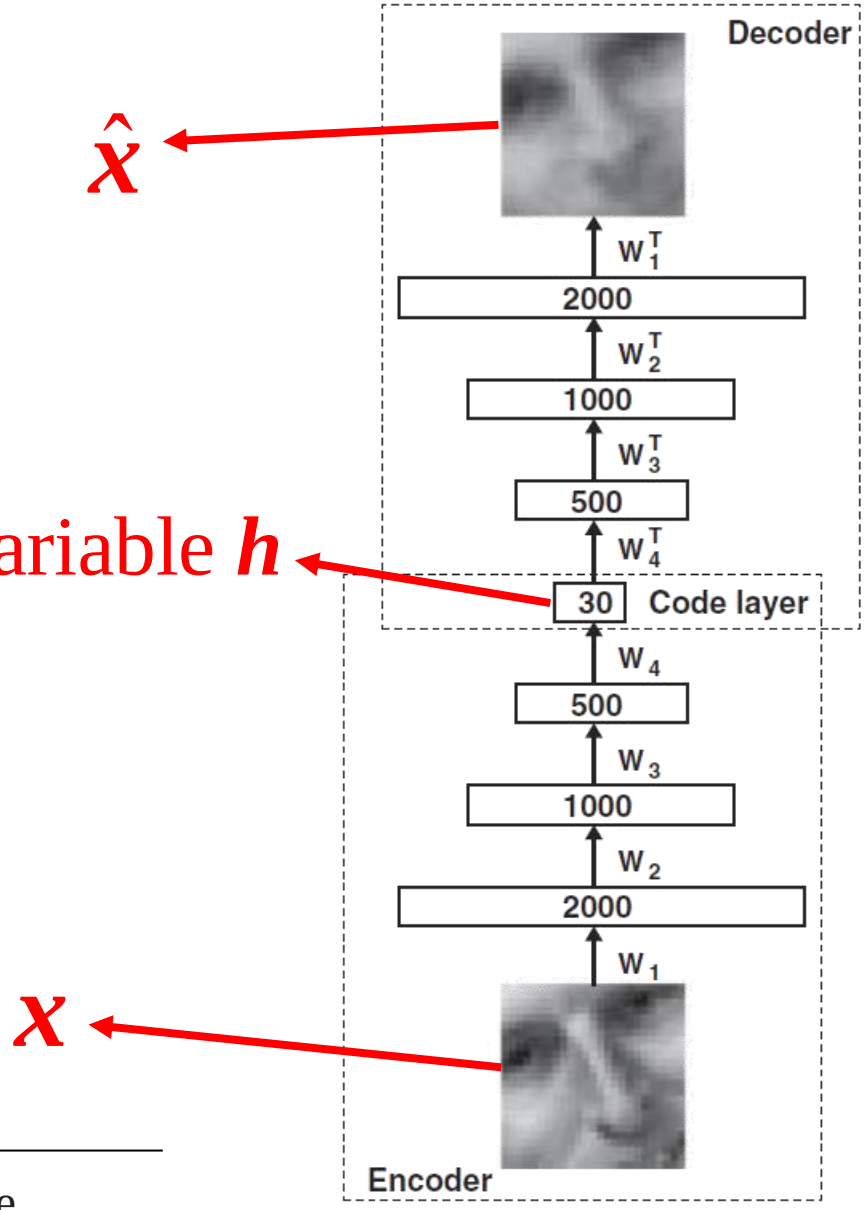
Autoencoder (AE)



Low dimensional hidden variable $\boldsymbol{h}$

Hinton, Geoffrey E., and Ruslan R. Salakhutdinov. "Reducing the dimensionality of data with neural networks." *science* 313.5786 (2006): 504-507.

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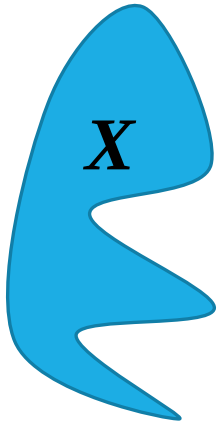
The Idea of Variational Autoencoder (VAE)

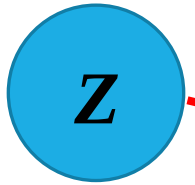


Observed Variable

Joint Density $p_{\theta*}(\boldsymbol{x}, \boldsymbol{z})$

$$p_{\theta*}(\boldsymbol{x}) = \int p_{\theta*}(\boldsymbol{x}, \boldsymbol{z})\mathrm{d}\boldsymbol{z} = \int p_{\theta*}(\boldsymbol{z})\boxed{p_{\theta*}(\boldsymbol{x} \mid \boldsymbol{z})}\mathrm{d}\boldsymbol{z}$$

Decoder

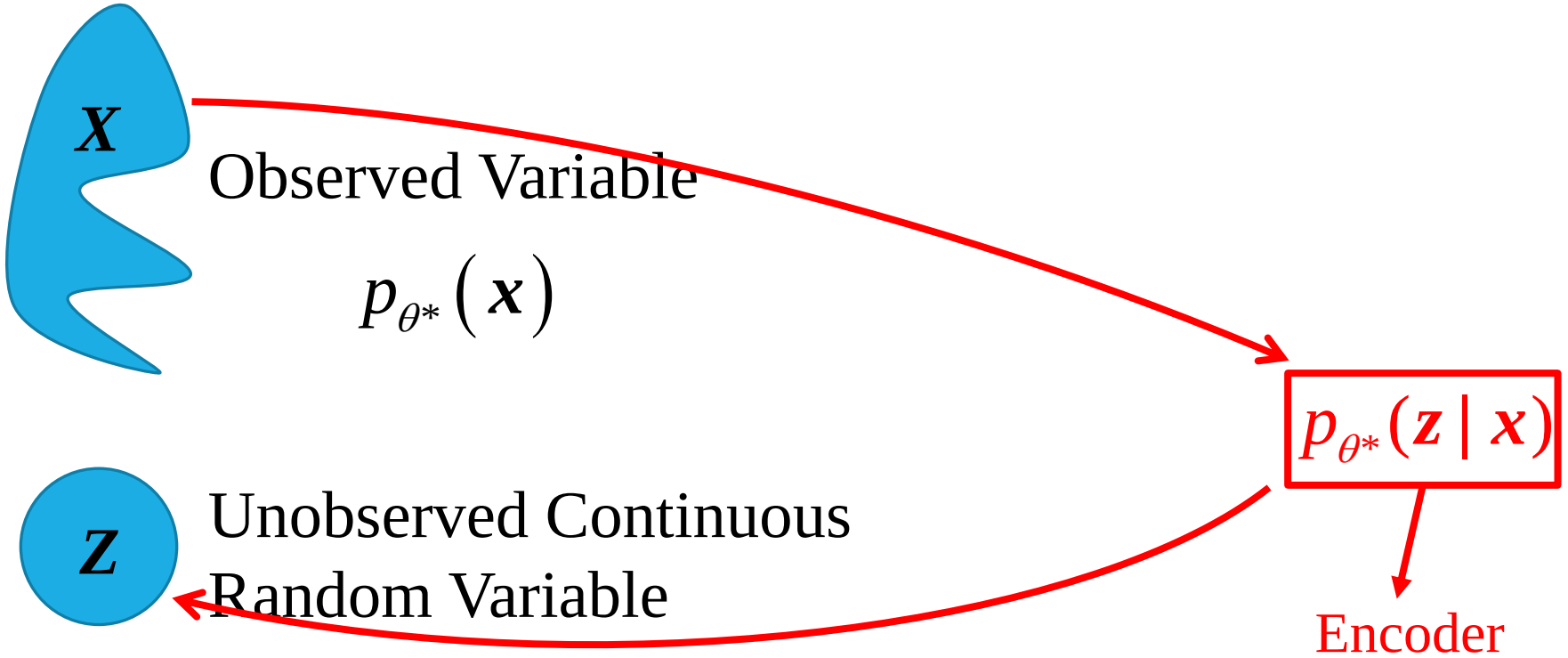


Unobserved Continuous Random Variable

$$p_{\theta*}(\boldsymbol{z}) \sim N(\boldsymbol{0}, \boldsymbol{I})$$

Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013).

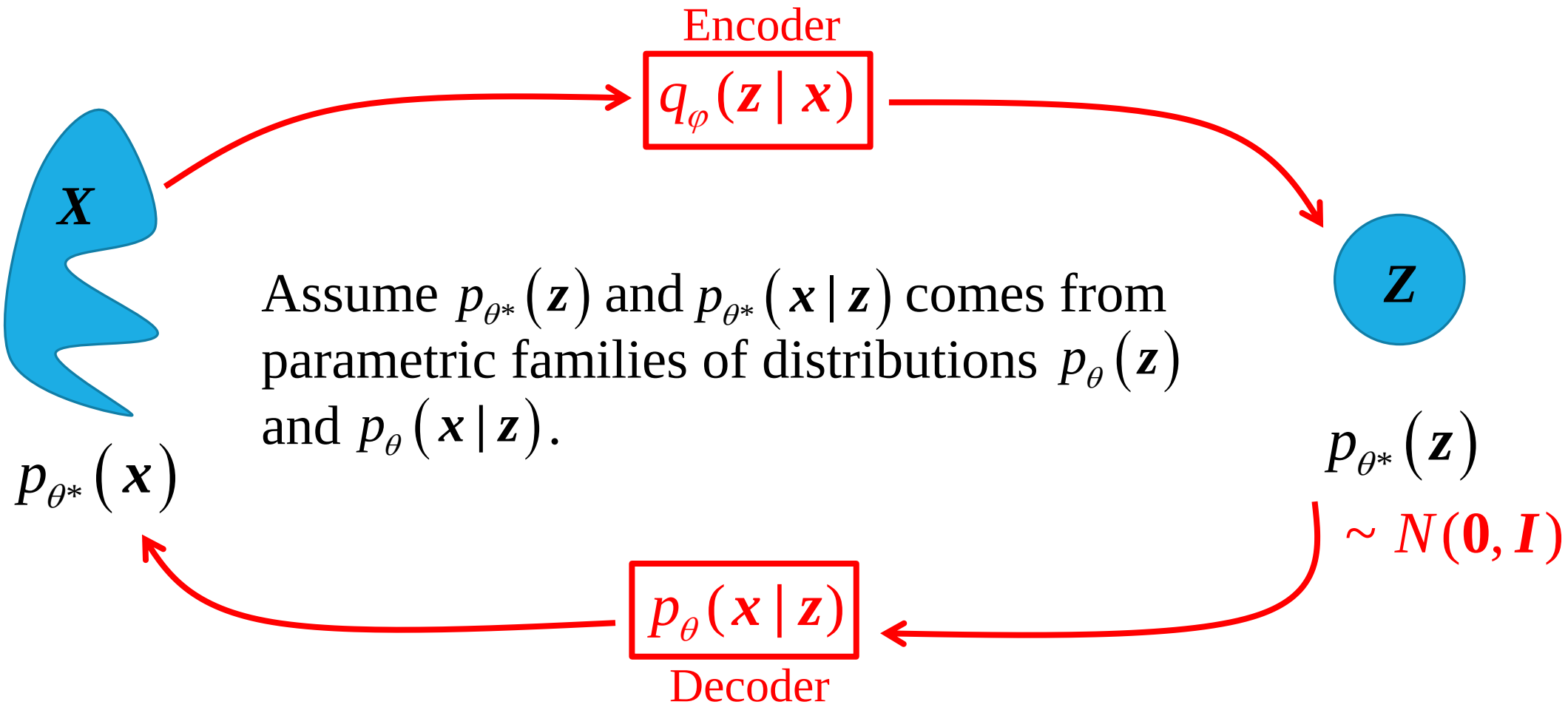
The Idea of Variational Autoencoder (VAE)



$$p_{\theta^*}(\boldsymbol{z}) = \int p_{\theta^*}(\boldsymbol{x}, \boldsymbol{z})\mathrm{d}\boldsymbol{x} = \int p_{\theta^*}(\boldsymbol{x})\, p_{\theta^*}(\boldsymbol{z} \mid \boldsymbol{x})\mathrm{d}\boldsymbol{x}$$

Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013).

The Idea of Variational Autoencoder (VAE)



Assume $p_{\theta^*}(\boldsymbol{z})$ and $p_{\theta^*}(\boldsymbol{x} \mid \boldsymbol{z})$ comes from parametric families of distributions $p_\theta(\boldsymbol{z})$ and $p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$.

Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013).

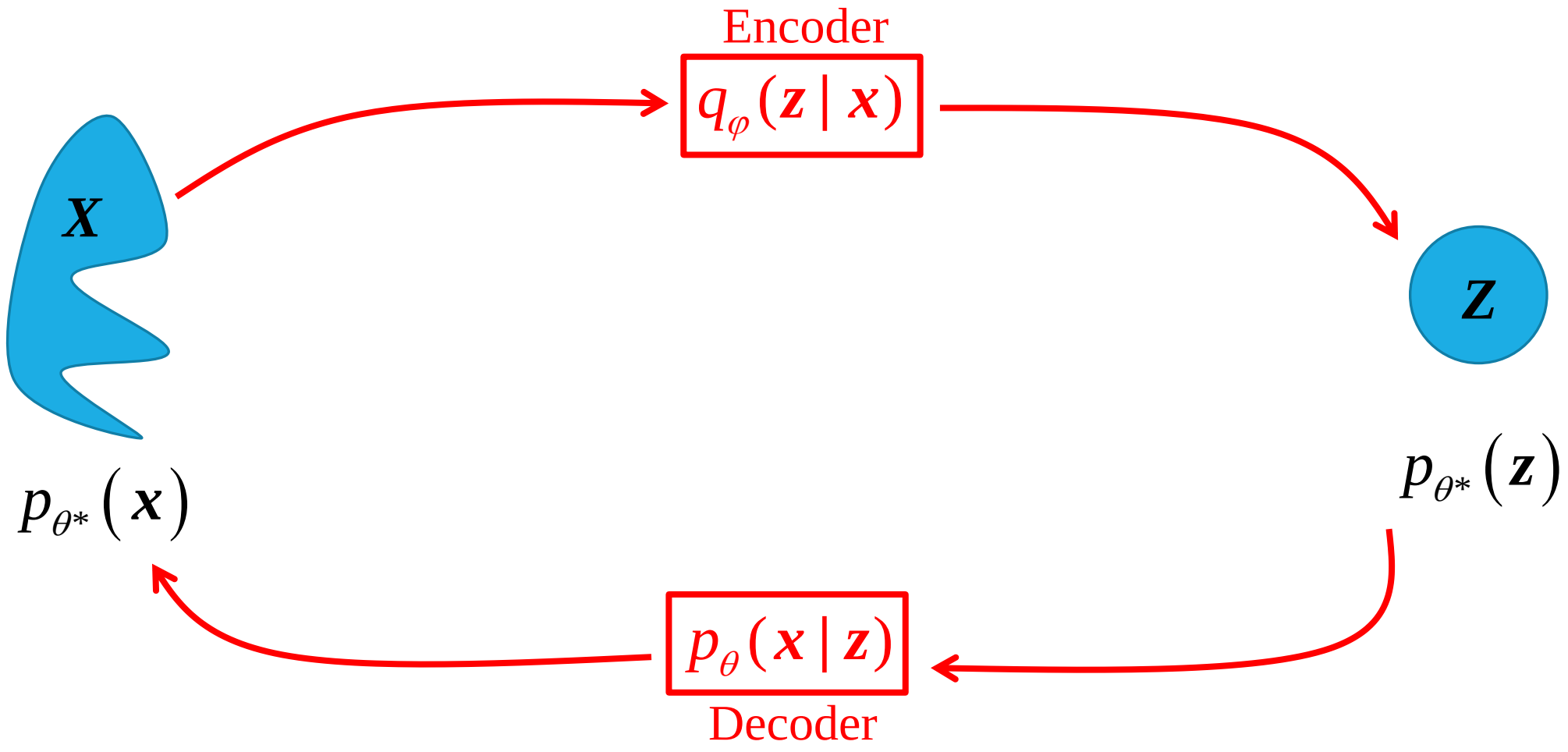
The Idea of Variational Autoencoder (VAE)

Maximum-likelihood

$$\int p_\theta(\boldsymbol{z}) p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) \mathrm{d}\boldsymbol{z} = p_\theta(\boldsymbol{x})$$

Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013).

The Idea of Variational Autoencoder (VAE)



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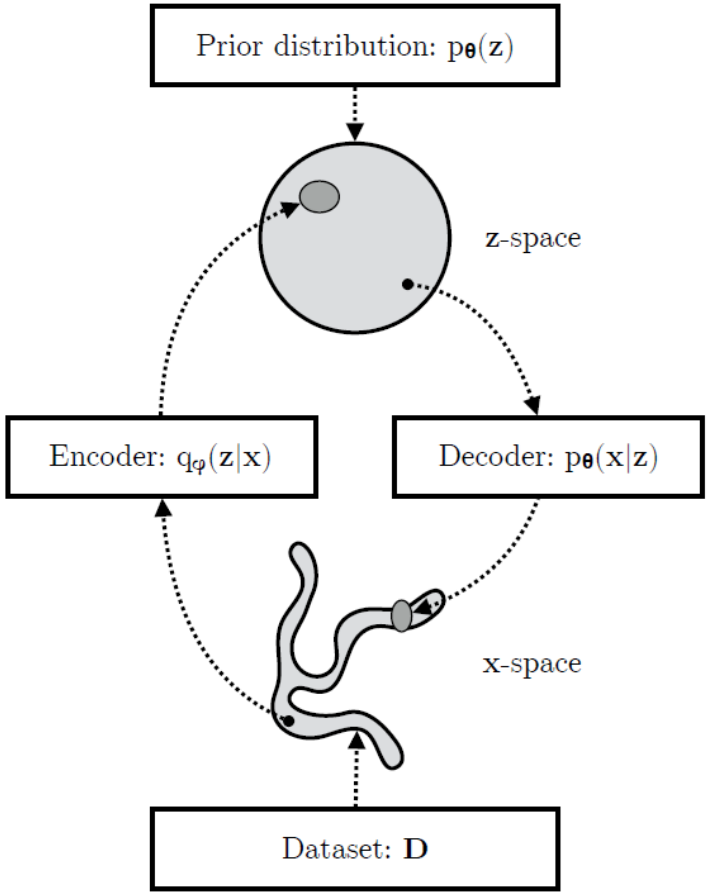
The Idea of Variational Autoencoder (VAE)



Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

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Variational Lower Bound

Maximum-likelihood

$$\int p_\theta(\boldsymbol{z}) p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) \mathrm{d}\boldsymbol{z} = p_\theta(\boldsymbol{x})$$

The dataset $\{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}\}$ consists of N i.i.d. samples.

The maximum-likelihood function are given by

$$\ln p_\theta(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}) = \ln \prod_{i=1}^{N} p_\theta(\boldsymbol{x}^{(i)}) = \sum_{i=1}^{N} \ln p_\theta(\boldsymbol{x}^{(i)})$$

$$\ln p_\theta(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x}, \boldsymbol{z})}{p_\theta(\boldsymbol{z} \mid \boldsymbol{x})}$$

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Variational Lower Bound

$$\ln p_\theta(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{p_\theta(\boldsymbol{z}\mid\boldsymbol{x})}$$

$$= \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})} \cdot \frac{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})}{p_\theta(\boldsymbol{z}\mid\boldsymbol{x})}$$

$$= \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})} + \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \frac{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})}{p_\theta(\boldsymbol{z}\mid\boldsymbol{x})}$$

$$= \mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) + D_{KL}(q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}\mid\boldsymbol{x})) \geq \mathcal{L}_{\theta,\varphi}(\boldsymbol{x})$$ Variational Lower Bound

$$D_{KL}(q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}\mid\boldsymbol{x})) \geq 0,$$

and the equality holds if and only if $q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) = p_\theta(\boldsymbol{z}\mid\boldsymbol{x})$

Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

Variational Lower Bound

$$\max_{\theta} \ln p_\theta(\boldsymbol{x}) \leftrightarrow \max_{\theta,\varphi} \mathcal{L}_{\theta,\varphi}(\boldsymbol{x})$$

$$\mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})} = \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{z})p_\theta(\boldsymbol{x}\mid\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})}$$

$$= \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x},\boldsymbol{z}) + \boxed{\boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})}}$$

$$= \boldsymbol{E}_{\boldsymbol{z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x}\mid\boldsymbol{z}) - D_{KL}(q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}))$$

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Variational Lower Bound

We can assume $p_\theta(\boldsymbol{x}\,|\,\boldsymbol{z}) \sim N(D_\theta(\boldsymbol{z}), \sigma^2 \boldsymbol{I})$, that is $\boldsymbol{X}$ satisfies Gaussian distribution $N(D_\theta(\boldsymbol{z}), \sigma^2 \boldsymbol{I})$ given that $\boldsymbol{Z} = \boldsymbol{z}$.

$$\mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x}\,|\,\boldsymbol{z}) - D_{KL}(q_\varphi(\boldsymbol{z}\,|\,\boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}))$$

$$\nabla_\theta \mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \nabla_\theta \ln p_\theta(\boldsymbol{x},\boldsymbol{z}) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \frac{1}{p_\theta(\boldsymbol{x},\boldsymbol{z})} \nabla_\theta p_\theta(\boldsymbol{x},\boldsymbol{z})$$

$$\nabla_\varphi \mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \nabla_\varphi \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\,|\,\boldsymbol{x})} \neq \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \nabla_\varphi \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\,|\,\boldsymbol{x})}$$

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Reparameterization Trick

$$\mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})}$$

$$\boldsymbol{Z} = \boldsymbol{\mu}(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim P(\boldsymbol{\varepsilon}),$$

$$\Sigma = Q\begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_{\mathrm{n}}^2 \end{bmatrix} Q^T = AA = AA^T$$

$$\mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{\varepsilon}\sim P(\boldsymbol{\varepsilon})} \ln p_\theta(\boldsymbol{x},\boldsymbol{z}) - \boldsymbol{E}_{\boldsymbol{\varepsilon}\sim P(\boldsymbol{\varepsilon})} \ln q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})$$

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Reparameterization Trick

Assume $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{I})$, then

$$\boldsymbol{Z} \sim q_{\varphi}(\boldsymbol{z} \mid \boldsymbol{x}) = N(\boldsymbol{\mu}(\boldsymbol{x}), \boldsymbol{\Sigma}(\boldsymbol{x})), \boldsymbol{\Sigma}(\boldsymbol{x}) = \boldsymbol{A}(\boldsymbol{x})\boldsymbol{A}(\boldsymbol{x})^{T}.$$

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Gaussian Distribution

Assume $\boldsymbol{\mu} \in \mathbb{R}^{n}$ and $\boldsymbol{\Sigma}$ is positive definite matrix of order n. Let $\boldsymbol{X}$ be a random vector of dimension n and the density function is given by

$$p(\boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{n}{2}}(\det \boldsymbol{\Sigma})^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\},\ \boldsymbol{x} \in \mathbb{R}^{n}.$$

We say $\boldsymbol{X}$ satisfies the Gaussian distribution.

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Variational Lower Bound (Evidence Lower Bound)

$$\mathcal{L}_{\theta,\varphi}(\boldsymbol{x}) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{x}\,|\,\boldsymbol{z}) - D_{KL}(q_\varphi(\boldsymbol{z}\,|\,\boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}))$$

Assume $p_\theta(\boldsymbol{x}\,|\,\boldsymbol{z}) \sim N(D_\theta(\boldsymbol{z}), \sigma^2 \boldsymbol{I})$,

$$p_\theta(\boldsymbol{x}\,|\,\boldsymbol{z}) = \frac{1}{(2\pi)^{\frac{n}{2}}\sigma^n} \exp\left\{-\frac{1}{2\sigma^2}\left(\boldsymbol{x} - D_\theta(\boldsymbol{z})\right)^T \left(\boldsymbol{x} - D_\theta(\boldsymbol{z})\right)\right\}$$

Then we have

$$\ln p_\theta(\boldsymbol{x},\boldsymbol{z}) = -\frac{1}{2\sigma_{\boldsymbol{z}}^2}\left\|\boldsymbol{x} - D_\theta(\boldsymbol{z})\right\|^2 - \frac{n}{2}\ln(2\pi\sigma_{\boldsymbol{z}}^2)$$

Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

Variational Lower Bound (Evidence Lower Bound)

$$\text{Maximum } \ln p_\theta(\boldsymbol{x} \mid \boldsymbol{z})$$

$$\leftrightarrow$$

$$\text{Minimum } \boldsymbol{E}_{\boldsymbol{z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \frac{1}{2\sigma_{\boldsymbol{z}}^2} \left\| \boldsymbol{x} - D_\theta(\boldsymbol{z}) \right\|^2 + D_{KL}(q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z}))$$

Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

Variational Lower Bound (Evidence Lower Bound)

$$D_{KL}(q_\varphi(\boldsymbol{z}\mid\boldsymbol{x})\,\|\,p_\theta(\boldsymbol{z})) = \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\ln q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) - \boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\ln p_\theta(\boldsymbol{z})$$

Assume $p_\theta(\boldsymbol{z}) = \dfrac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\|\boldsymbol{z}\|^2}$, and

$$q_\varphi(\boldsymbol{z}\mid\boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{n}{2}}(\det\boldsymbol{\Sigma})^{\frac{1}{2}}}\exp\left\{-\frac{1}{2}\left(\boldsymbol{z}-\boldsymbol{\mu}\right)^T\boldsymbol{\Sigma}^{-1}\left(\boldsymbol{z}-\boldsymbol{\mu}\right)\right\},$$

where $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^T, \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{pmatrix}.$

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Variational Lower Bound (Evidence Lower Bound)

$$D_{KL}(q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z})) = \boldsymbol{E}_{\boldsymbol{Z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) - \boldsymbol{E}_{\boldsymbol{Z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{z})$$

$$q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} \prod_{i=1}^{n} \sigma_i} \exp\left\{ -\frac{1}{2} \sum_{i=1}^{n} \frac{(z_i - \mu_i)^2}{\sigma_i^2} \right\},$$

$$\ln q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) = -\frac{n}{2} \ln 2\pi - \frac{1}{2} \sum_{i=1}^{n} \ln \sigma_i^2 - \frac{1}{2} \sum_{i=1}^{n} \frac{(z_i - \mu_i)^2}{\sigma_i^2},$$

Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

Variational Lower Bound (Evidence Lower Bound)

$$D_{KL}(q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z})) = \boldsymbol{E}_{\boldsymbol{Z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) - \boldsymbol{E}_{\boldsymbol{Z} \sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})} \ln p_\theta(\boldsymbol{z})$$

$$q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} \prod_{i=1}^{n} \sigma_i} \exp\left\{ -\frac{1}{2} \sum_{i=1}^{n} \frac{(z_i - \mu_i)^2}{\sigma_i^2} \right\},$$

$$\int_{\mathbb{R}^n} \frac{(z_i - \mu_i)^2}{\sigma_i^2} q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) \mathrm{d}\boldsymbol{z} = \int_{-\infty}^{+\infty} \frac{(z_i - \mu_i)^2}{\sigma_i^2} \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(z_i - \mu_i)^2}{2\sigma_i^2}} \mathrm{d}z_i$$

$$= \int_{-\infty}^{+\infty} t^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \mathrm{d}t = 1, \text{where } t = \frac{z_i - \mu_i}{\sigma_i}$$

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Variational Lower Bound (Evidence Lower Bound)

$$\ln q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\sum_{i=1}^{n}\ln \sigma_i^2 - \frac{1}{2}\sum_{i=1}^{n}\frac{(z_i-\mu_i)^2}{\sigma_i^2},$$

$$\boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\ln q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\sum_{i=1}^{n}\left(1+\ln \sigma_i^2\right)$$

Similarly, $\boldsymbol{E}_{\boldsymbol{Z}\sim q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\ln p_\theta(\boldsymbol{z}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\sum_{i=1}^{n}\left(\mu_i^2+\sigma_i^2\right).$

$$D_{KL}(q_\varphi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z})) = \frac{1}{2}\sum_{i=1}^{n}\left(\mu_i^2+\sigma_i^2-1-\ln \sigma_i^2\right)$$

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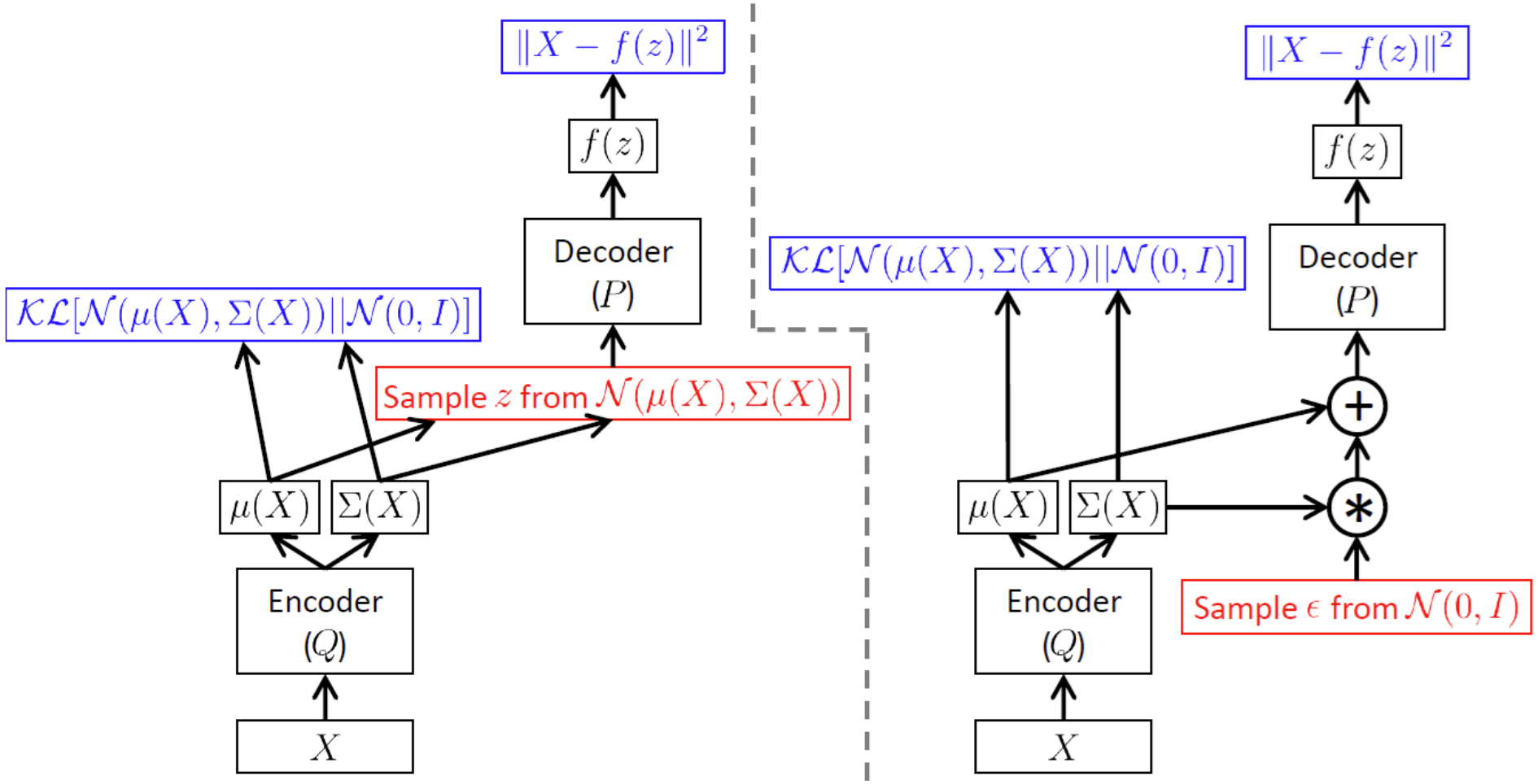
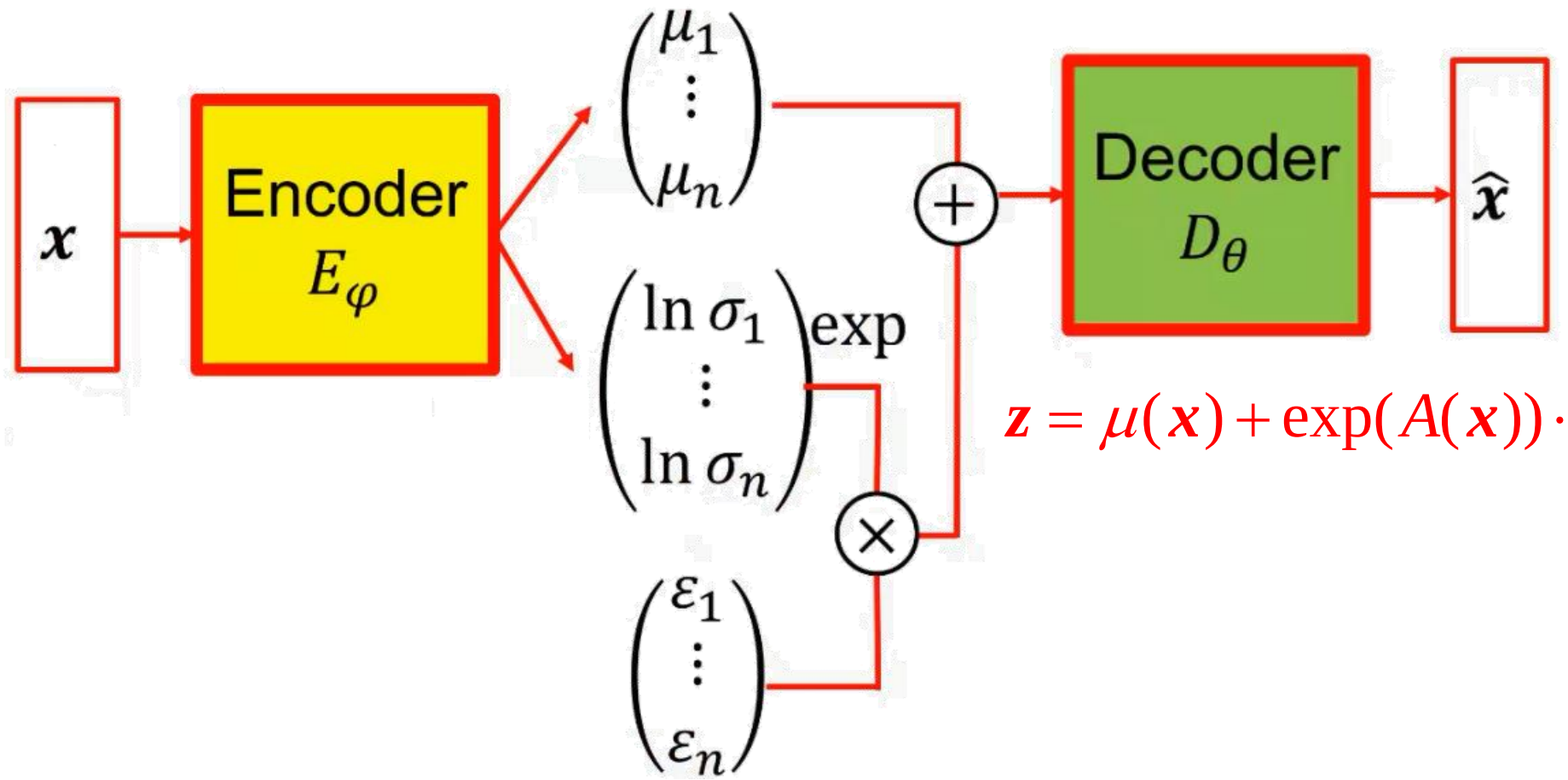


Doersch, Carl. "Tutorial on variational autoencoders." *arXiv preprint arXiv:1606.05908* (2016).

VAE Network



$$\boldsymbol{z} = \mu(\boldsymbol{x}) + \exp(A(\boldsymbol{x})) \cdot \boldsymbol{\varepsilon}$$

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REFERENCES (F.Y.I.)

[1] Hinton, Geoffrey E., and Ruslan R. Salakhutdinov. "Reducing the dimensionality of data with neural networks." *science* 313.5786 (2006): 504-507.

[2] Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013).

[3] Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

[4] Doersch, Carl. "Tutorial on variational autoencoders." *arXiv preprint arXiv:1606.05908* (2016).

[5] Hung-Yi Lee, https://speech.ee.ntu.edu.tw/~hylee/ml/2021-spring.php.

THANK YOU!

XU Chentianye · 20 Nov, 2021